

DDL, DML & TCL Commands

Think of a Database as a House

Imagine a Database as a House. DDL builds the house, DML manages the things inside, and TCL controls whether changes are saved or cancelled.

1. DDL – Data Definition Language

“Building the Database”

DDL is used to create and modify the structure of database objects such as tables.

Command	What it does	Simple idea
CREATE	Creates a new table/database	Build
ALTER	Changes the table structure	Modify
DROP	Removes the table/database	Remove
TRUNCATE	Removes all records from a table	Empty

Example:

```
CREATE TABLE Student (  
  id INT,  
  name VARCHAR(50)  
);
```

Memory: DDL = Design the Database Layout

2. DML – Data Manipulation Language

“Managing the Data Inside”

DML is used to add, modify and remove records stored inside a table.

Command	What it does	Simple idea
INSERT	Adds new records	Add
UPDATE	Modifies existing records	Edit
DELETE	Removes records	Remove

Example:

```
INSERT INTO Student VALUES (1, 'Madhu');
```

```
UPDATE Student  
SET name = 'Madhumitha'  
WHERE id = 1;
```

```
DELETE FROM Student  
WHERE id = 1;
```

Memory: DML = Deal with the Data

3. TCL – Transaction Control Language

“Controlling the Changes”

TCL is used to manage database transactions and decide whether changes should be saved or undone.

Command	What it does	Simple idea
COMMIT	Permanently saves changes	SAVE
ROLLBACK	Cancels changes after the last commit	UNDO
SAVEPOINT	Creates a point to which we can return	CHECKPOINT

Example:

```
INSERT INTO Student VALUES (1, 'Madhu');
```

```
SAVEPOINT s1;
```

```
UPDATE Student
SET name = 'Madhumitha'
WHERE id = 1;
```

```
ROLLBACK TO s1;
```

```
COMMIT;
```

Memory: TCL = Take Control of Changes

The “House” Trick

DDL → Build the House

CREATE → Build | ALTER → Modify | DROP → Demolish | TRUNCATE → Empty

DML → Manage the Things

INSERT → Add | UPDATE → Change | DELETE → Remove

TCL → Control Your Decisions

COMMIT → Save | ROLLBACK → Undo | SAVEPOINT → Checkpoint

Super-short memory line: DDL = BUILD → DML = MANAGE → TCL = CONTROL